

NEET Important Topic – Physics Mathematical Tools

Physics Mathematical Tools for NEET

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As far as the NEET aspirants are concerned, one of the challenges they face while studying physics for the NEET exam is the mathematical operations and tools used at the time of solving numerical problems. However, this can be made easier if we learn the basic mathematical tools required in physics for the NEET exam.

In this article discussing the new and old mathematical tools in physics, we will see different physical constants and conversion coefficients that will come in handy while solving numerical problems. The importance of mathematics of tools like trigonometric functions of angles and their applications are discussed here. The application of differentiation and integration plays an important role in solving most of the physics problems in several chapters.

Now, let us move on to the important concepts and formulae of mathematical tools required in physics for the NEET exam, along with a few solved mathematical tools examples.

Important Topics of Mathematical Tools

- Quadratic equation
- Binomial expansion
- Important trigonometric identities.
- Differentiation
- Rules of differentiation
- Applications of differentiation
- Integration and its application
- Definite integral
- Integration Formulae

Important Concepts of Mathematical Tools

Name of the Concept	Key Points of Concept
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1. Quadratic equation

- A quadratic equation is a second-degree equation in which the variable is raised to the power 2.
- The quadratic equation form is given below where a , b and c are constants.
- The discriminant of the quadratic equation is given by,

$$b^2 - 4ac$$
- The solution of the quadratic equation is given by,

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$
- One of the applications of quadratic equations in physics is calculating the time taken for a ball to reach a certain height H . Here, we will get two values for time, t_1 and t_2 , after solving the quadratic equations.

2. Binomial expansion	• The pattern of binomial expansion is given by the binomial theorem _____ • Similarly, the expansion of $(1+x)^n$ is given by, _____ • If the values x is very small, we can neglect the higher powers of x in binomial expansion as
3. Important trigonometric identities.	• Trigonometric ratios has positive or negative values depending on the quadrant given below

Image: Trigonometric function signs

- Some of the important trigonometric relations are given below

- Let y be a function of x , then ratio of change of y for a very small change in x is represented by

- One of the applications of differentiation is finding the slope of y

4. Differentiation

Image: application of differentiation

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- Some of the differentiation of basic function is given below.

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5. Rules of differentiation

- Let $f(x)$ and $g(x)$ be two functions of x . The addition and subtraction differentiation is given by

- Constant rule:

Where k is a constant

- Product rule:

- Quotient rule:

- Velocity is the rate of change of displacement with respect to t
—
- The slope of the displacement time graph gives velocity.

6.
Applications
of
differentiation

Image: slope of displacement time graph

- Acceleration is the rate of change of velocity with respect to time

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Image: slope of velocity time graph

- The slope of the velocity time graph gives the acceleration.

- Integration is basically the summation of infinite quantities, of i
- If $f(x)$ is the derivative of $g(x)$ with respect to x , then $g(x)$ is the i

—

Where c is an arbitrary constant.

- We can calculate the area under the curve of a given graph us

7. Integration
and its
application

Image: Area under the curve

- Some of integral of basic function is given below

$$\int \frac{1}{x} dx = \ln|x| + C$$

8. Definite
integration

- When an integral is defined between two definite limits a and b definite integration.

List of Important Formulae

Sl. No	Name of the Concept	Formula
1.	Trigonometric formulae	
2.	Differentiation of trigonometric functions	— — — —
3.	Logarithmic formulae	—
4.	Integration formulae of trigonometric function	
5.	Conversion coefficients	

Solved Examples

1. The position of a particle on x-axis that varies with time is given as $x=3t^2+2t+1$. Find the velocity of the particle at $t=2$ s.

Sol:

Given,

The position of the particle, $x=3t^2+2t+1$

The velocity of particle at $t=2$ s can be found by differentiating the displacement and substituting the value of t as 2 sec

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—

Put $t=2$ to calculate the velocity at that instant.

Therefore, the velocity of the particle at $t=2$ s is 14 m/s. **Key point:** Differentiating displacement with respect to time gives velocity.

2. The variable force acting on a body of mass 1 kg depends on the displacement on x -axis as $F=3x^2+2$. Find the work done by the variable force from $x=0$ to $x= 2$ m.

Sol:

Given,

The variable force acting on the body, $F=3x^2+2$

We can calculate the work done by the force by integrating force from $x=0$ m to $x= 2$ m

—

Thus, the work done by the variable force from $x=0$ to $x= 2$ m is 12J.

Key Point

: The work done by a variable force can be obtained by integration of the variable force with respect to displacement.

Previous Year Questions from NEET Paper

1. A particle moves along a straight line OX. At a time t (in seconds), the distance x (in metres) is given by $x=40+12t-t^3$. How long would the particle travel before coming to rest? (NEET 2006)

a. 24 m

b. 40 m

c. 56 m

d. 16 m

Sol:

The speed of the particle can be obtained by differentiating the distance with respect to time.

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—

When particle comes to rest, the speed of the particle becomes zero

Now, the distance travelled by the particle is

Therefore, correct answer is option (c)

Trick: Speed of the particle can be obtained by differentiating the expression for distance with respect to time.

2. The displacement of a particle moving in a straight line is given by $s=2t^2+2t+4$, where s is in metres and t is in seconds. The acceleration of the particle is (NEET 2001)

a. 2 m/s^2

b. 4 m/s^2

c. 6 m/s^2

d. 8 m/s^2

Sol:

The speed of the particle can be obtained by

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The acceleration of the particle is calculated as follows

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Therefore, correct answer is option (b)

Trick: When displacement of any object is given second-order differentiation will be the acceleration of the object.

Practice Questions

1. Find the work done to elongate a spring having a spring constant of 2000 N/m from an extension of 1 cm to 2 cm .

Ans: 0.3 J

2. If $v=x$

$^2-5x+4$, find the acceleration of the particle when the velocity of the particle is zero.

Ans: 0 m/s^2

Conclusion

In this article, we have provided important information under several mathematical tool names necessary for solving numerical problems in Physics. Students should have a better understanding of the mathematical operations so that students can score good marks in Physics for the NEET exam.

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